

# STAT 542 - Statistical Learning

## Homework Two

Fall 2026, University of Illinois Urbana-Champaign

Due date: TBA  
Total Points = 75

1. We use the following fact:

$$\begin{aligned} \theta \sim N(\mu, \tau^2) \quad \text{and} \quad y \mid \theta \sim N(\theta, \sigma^2) &\implies \theta \mid y \sim N\left(\frac{y/\sigma^2 + \mu/\tau^2}{1/\sigma^2 + 1/\tau^2}, \frac{1}{1/\sigma^2 + 1/\tau^2}\right) \\ &\text{and } y \sim N(\mu, \sigma^2 + \tau^2). \end{aligned} \tag{1}$$

The goal of this problem is to prove a much more general form of this result (this general form will be useful later on in the course).

Suppose  $X$  and  $Y$  are random vectors with  $X \sim N(m_0, Q_0)$  and  $Y \mid X = x \sim N(Bx, R)$ . Let  $X \mid Y = y \sim N(m_1, Q_1)$ .

- a) Show that (**4 points**)

$$\begin{aligned} m_1 &= (I + Q_0 B' R^{-1} B)^{-1} m_0 + Q_0 B' R^{-1} y, \\ Q_1 &= (I + Q_0 B' R^{-1} B)^{-1} Q_0. \end{aligned}$$

- b) Show that (**3 points**)

$$\begin{aligned} m_1 &= m_0 + Q_0 B' (B Q_0 B' + R)^{-1} (y - B m_0), \\ Q_1 &= Q_0 - Q_0 B' (B Q_0 B' + R)^{-1} B Q_0. \end{aligned}$$

- c) Show that (**2 points**)

$$Y \sim N(B m_0, B Q_0 B' + R).$$

- d) Deduce (1) as a consequence of the results in parts (a)–(c). (**2 points**)

2. Consider the multiple regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_m x_{im} + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2). \tag{2}$$

We observe data  $(x_{i1}, \dots, x_{im}, y_i)$ ,  $i = 1, \dots, n$ , and treat the covariates as fixed constants. For Bayesian analysis use the prior

$$\beta_0, \beta_1, \dots, \beta_m, \log \sigma \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(-C, C).$$

As  $C \rightarrow \infty$ ,

$$\beta \mid \text{data} \sim t_{n-m-1, m+1} \left( \hat{\beta}, \hat{\sigma}^2 (X^T X)^{-1} \right), \quad (3)$$

where  $X$  is the design matrix,  $\beta$  is the coefficient vector,  $\hat{\beta}$  is the least-squares estimator, and  $\hat{\sigma}$  is the usual residual standard-error estimator. This exercise gives another proof of the same result and discusses inference for  $\sigma$ .

a) Prove that

$$\beta \mid \text{data}, \sigma \sim N_{m+1} \left( \hat{\beta}, \sigma^2 (X^T X)^{-1} \right). \quad (4)$$

The left-hand side is the conditional density of  $\beta$  given the data and the parameter  $\sigma$ . (**4 points**)

b) Prove that the posterior density of  $\sigma$  is proportional to

$$\sigma^{-n+m} \exp \left( -\frac{S(\hat{\beta})}{2\sigma^2} \right) I\{\sigma > 0\}, \quad (5)$$

where  $S(\hat{\beta})$  is the residual sum of squares. (**4 points**)

c) Use the results of the previous two parts and

$$f_{\beta \mid \text{data}}(\beta) = \int_0^\infty f_{\beta \mid \text{data}, \sigma}(\beta) f_{\sigma \mid \text{data}}(\sigma) d\sigma$$

to deduce (3). (**4 points**)

d) Use the result in part (b) to show that

$$\frac{S(\hat{\beta})}{\sigma^2} \mid \text{data} \sim \chi_{n-m-1}^2.$$

(**3 points**)

e) Use the result in the previous part to argue that the usual estimator

$$\hat{\sigma} := \sqrt{\frac{S(\hat{\beta})}{n-m-1}}$$

is a reasonable point estimate of  $\sigma$ . (**2 points**)

3. This problem discusses predictions in linear regression. Consider the same setting as in the previous problem.

a) Suppose we are interested in the parameter  $a^T \beta$  for some fixed vector  $a$ . Using (4), prove that

$$a^T \beta \mid \text{data}, \sigma \sim N \left( a^T \hat{\beta}, \sigma^2 a^T (X^T X)^{-1} a \right).$$

Note that we condition on the data and on  $\sigma$ . (**3 points**)

b) Using (3), prove that

$$a^T \beta \mid \text{data} \sim t_{n-m-1} \left( a^T \hat{\beta}, \hat{\sigma}^2 a^T (X^T X)^{-1} a \right).$$

(**3 points**)

c) Suppose we want to predict  $Y_{n+1}$  at new covariates  $x_{n+1,1}, \dots, x_{n+1,m}$ . Assuming

$$Y_{n+1} = \beta_0 + \beta_1 x_{n+1,1} + \dots + \beta_m x_{n+1,m} + \epsilon_{n+1},$$

where  $\epsilon_{n+1} \sim N(0, \sigma^2)$  is independent, prove that

$$Y_{n+1} \mid \text{data}, \sigma \sim N\left(\tilde{x}_{n+1}^T \hat{\beta}, \sigma^2 + \sigma^2 \tilde{x}_{n+1}^T (X^T X)^{-1} \tilde{x}_{n+1}\right),$$

where  $\tilde{x}_{n+1} = (1, x_{n+1,1}, \dots, x_{n+1,m})^T$ . ( **4 points** )

d) Using the result from the previous part, the formula

$$f_{Y_{n+1} \mid \text{data}}(y) = \int_0^\infty f_{Y_{n+1} \mid \text{data}, \sigma}(y) f_{\sigma \mid \text{data}}(\sigma) d\sigma,$$

and the density  $\sigma \mid \text{data}$  from (5), prove that

$$Y_{n+1} \mid \text{data} \sim t_{n-m-1}\left(\tilde{x}_{n+1}^T \hat{\beta}, \hat{\sigma}^2 + \hat{\sigma}^2 \tilde{x}_{n+1}^T (X^T X)^{-1} \tilde{x}_{n+1}\right).$$

This result can be used to obtain predictions and uncertainty intervals. ( **5 points** )

4. Consider the `spam7.csv` dataset, which consists of data on  $n = 4601$  emails corresponding to the following variables:

- **cr1.tot**: Total length of uninterrupted sequences of capital letters.
- **dollar**: Occurrences of the dollar sign (\$), expressed as a percentage of the total number of characters.
- **bang**: Occurrences of the symbol !, expressed as a percentage of the total number of characters.
- **money**: Occurrences of the word “money”, expressed as a percentage of the total number of words.
- **n000**: Occurrences of the string “000”, expressed as a percentage of the total number of words.
- **make**: Occurrences of the word “make”, expressed as a percentage of the total number of words.
- **yesno**: Indicates whether the message is spam, with levels

$$n = \text{not spam}, \quad y = \text{spam}.$$

Our goal is to perform logistic regression on this dataset without using any existing libraries. Our response variable  $y$  will be the **yesno** variable, which indicates whether the email is spam or not. As covariates, use the following six variables:

$$\begin{aligned} x_1 &= \log(\text{cr1.tot}), & x_2 &= \log(\text{dollar} + 0.001), & x_3 &= \log(\text{bang} + 0.001), \\ x_4 &= \log(\text{money} + 0.001), & x_5 &= \log(\text{n000} + 0.001), & x_6 &= \log(\text{make} + 0.001). \end{aligned}$$

We take logarithms because these variables are otherwise highly skewed. We add 0.001 because some of these variables can take the value zero and we do not want  $\log(0)$ .

- a) Using Newton's algorithm, calculate the maximum likelihood estimate of the coefficients in logistic regression. Check the correctness of your solution using an inbuilt library function for fitting logistic regression. (**4 points**)
- b) Compute the Laplace posterior normal approximation corresponding to the prior

$$\beta_0, \dots, \beta_m \stackrel{\text{i.i.d.}}{\sim} \text{uniform}(-\infty, \infty).$$

Compute the posterior standard errors for the coefficients. Check the correctness of your answers using an inbuilt library function for fitting logistic regression. (**4 points**)

5. The **probit regression model** is given as follows. We observe binary responses  $y_i$  and covariate vectors  $x_i \in \mathbb{R}^p$  for  $i = 1, \dots, n$ . The  $x_i$  are treated as non-random and the  $y_i$  as random. The probit regression model assumes that  $y_1, \dots, y_n$  are independent with

$$y_i \sim \text{Bernoulli}(\Phi(x_i' \beta)),$$

where  $\Phi(\cdot)$  is the standard normal cdf. The only difference from the logistic regression model is that  $\exp(x_i' \beta) / (1 + \exp(x_i' \beta))$  is replaced by  $\Phi(x_i' \beta)$ .

- a) Write down the log-likelihood for this model and calculate its gradient and Hessian for a generic parameter vector  $\beta$ . (**3 + 3 + 3 = 9 points**)
- b) Let  $\hat{\beta}$  be the MLE. Consider the prior  $\beta_j \stackrel{\text{i.i.d.}}{\sim} \text{uniform}(-\infty, \infty)$ . Express the posterior normal approximation in terms of  $\hat{\beta}$ . (**3 points**)
- c) Explicitly calculate the Laplace posterior normal approximation for the *frogs* dataset. Use the response and covariate variables specified for that dataset. You may use standard library functions to calculate the MLE  $\hat{\beta}$  or use Newton's method directly. (**3 points**)
- d) Compare the posterior standard deviations (as calculated from the posterior normal approximation) with the standard errors reported by a library function performing a standard frequentist analysis of probit regression. Are they similar? (**3 points**)

6. Consider the linear regression model

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

for  $i = 1, \dots, n$ , with  $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \text{Lap}(0, \sigma)$ . Recall that  $\text{Lap}(0, \sigma)$  is the Laplace or double exponential density

$$x \mapsto \frac{1}{2\sigma} \exp\left(-\frac{|x|}{\sigma}\right).$$

Consider the usual prior

$$\beta_0, \beta_1, \log \sigma \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(-\infty, \infty).$$

The goal is to fit this model to  $(x_1, y_1), \dots, (x_n, y_n)$ .

- a) Prove that (**3 points**)

$$f_{\beta_0, \beta_1 | \text{data}}(\beta_0, \beta_1) \propto \left( \frac{1}{S(\beta_0, \beta_1)} \right)^n, \quad (6)$$

where

$$S(\beta_0, \beta_1) := \sum_{i=1}^n |y_i - \beta_0 - \beta_1 x_i|.$$

- b) Consider the Pearson father-son height dataset. Fit this model to the data and report point estimates and standard errors of  $\beta_0$  and  $\beta_1$ . You may approximate the joint density (6) by a discrete distribution over a dense grid of  $\beta_0$  and  $\beta_1$ . (**5 points**)
- c) Compare your estimates of the intercept and slope with the corresponding estimates obtained by fitting the usual regression model with normal errors. Comment on any discrepancies (or similarities) between the two sets of results. (**2 points**)